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Lab4: Ecological Dynamics
Due by Monday April 27th, 2026 @ 11:59pm

The goal of this lab is to develop an agent-based model of predator-prey dynamics. Our goal is to understand how functional forms attributed to "handling times" arise from individual dynamics.

1 Functional responses - a microscopic approach

1.1 Theoretical background

Predators or consumers must interact with their prey or resource to consume it. Here, we will consider the "microscopic" basis for the kinds of functional forms used to described predator-prey and consumer-resource models. Recall predator-prey model:

$$\dot{N} = rN(1 - N/K) - bNP$$
$$\dot{P} = cNP - mP$$

where the term bNP denotes consumption of prey by predators. This form suggests a collision-based approach, but does it also apply to spatial dynamics? Moreover, how would this kind of functional form changed if predators had to "handle" hosts before consuming them. This lab and the homework will help you understand these questions.

First, let's assume that a predator moves "ballistically" through an environment with a velocity v. Moreover, assume it can sense prey in a zone of radius r. Hence, the predator will sense an area of approximate size $\pi r^2 + 2rvT_s$ in a period T_s of searching. If prey are at a density N, then the potential number of prey encounters is $(\pi r^2 + 2rvT_s)N$. Further if a fraction k are detected and of those f are consumed per unit time, then the total prey consumption is $(\pi r^2 + 2rvT_s)fkT_sN$ per predator in a period of time T_s. This is true for all predators, as long as they tend *not* to run into or near each other. Hence, I claim that the total consumption rate scales like

$$\text{consumption rate} = bNP$$

where the constant b approaches

$$\pi r^2 fk$$

How true is any of this? For example, you may find that this limit poses some problems, e.g., an immobile predator with relatively slow diffusing prey may rapidly deplete local prey. So even though it continues to sense across a radius of r, there may not be any prey left to sense! To test these ideas, I developed an individual based model (IBM) of predator detecting and eating prey. This code is yours to modify and manipulate.

1.2 Getting Started

First, download three files from Bruinlearn Lab4 Assignment page (should be in one zip file):

- `master_ibm.m` – the master script for setting parameters of the IBM model
- `ibm_predation.m` – dynamic simulator of predation
- `restore_pos.m` – utility script to move positions of particles back into a square domain

Once you've downloaded the files, go to matlab and at the prompt type:

```
>> master_ibm
```

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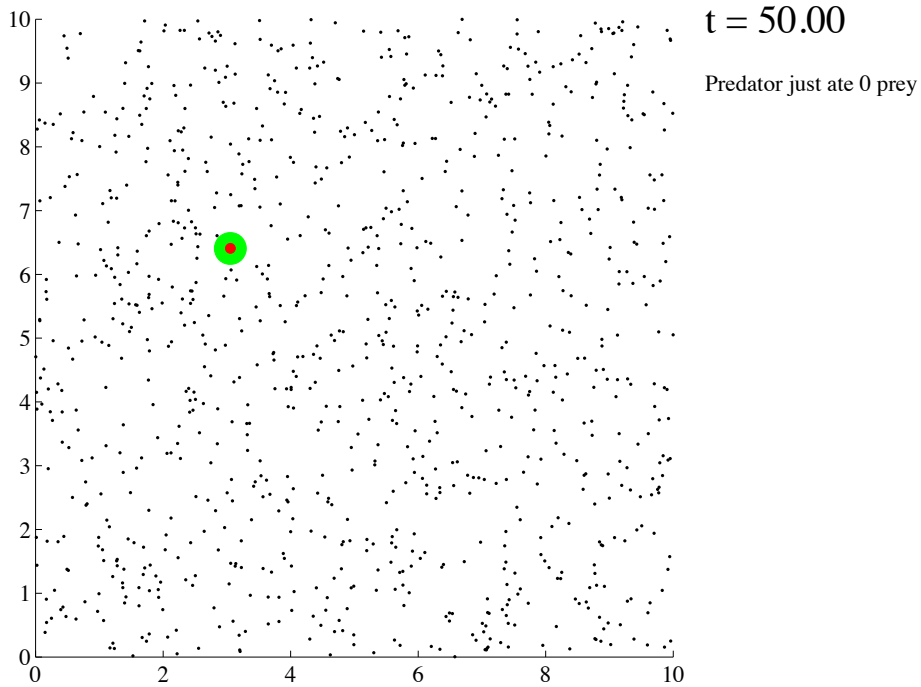
You should see a red predator with a green zone of detection eating prey. The predator moves ballistically (in a straight line) and changes directions on occasion, whereas the prey move diffusively. Blue circles denote the last set of prey eaten in a particular interval. They automatically update as new prey are consumed. Re-size the window so you can see the update label. This is what it should look like when done:



When your simulation is done, type:

```
>> figure; plot(t,total_eaten);
```

which will give you a display of the cumulative number of prey eaten by the predator with time. Now that the code is working, let's take a closer look:

```
% An individual-based model of predators eating prey
% Joshua Weitz

% Basic information
info.prey_density=10;   % density/cm^2
info.maxX=10;           % cm
info.maxY=10;           % cm
info.tf=50;             % sec
info.dt=0.1;            % Time step
info.replenish_prey=1;  % Should prey regenerate?
info.viz_dyn=0;  % 1 for animation, 0 for no animation

% Define the prey
prey.num=info.prey_density*info.maxX*info.maxY;
prey.pos=rand(prey.num,2);   % Random positions
prey.pos(:,1)=prey.pos(:,1)*info.maxX;
prey.pos(:,2)=prey.pos(:,2)*info.maxY;
prey.diffusion=0.005;   % cm^2/sec

% Place the predator
predator.pos=[info.maxX/2 info.maxY/2];
predator.theta=rand*2*pi;  % Angle of movement
```

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```
predator.r=1.25;            % Radius
predator.k=0.3;             % Detection
predator.f=0.05;             %  Successful capture
predator.vel=0.1;           % cm/sec
predator.handling_time=0.0;   % seconds
predator.tau=6;             % Run time length, seconds
predator.trun=0;            % Initialize each run

% Simulate eating
[t,numeaten]=ibm_predation(info,predator,prey);
total_eaten=cumsum(numeaten);
```

Do you understand what each of the terms mean? Also spend 5 minutes looking over the somewhat difficult function: ibm_predation.m.

Exercises 1 (75 pts): Exploration of predation and functional responses. For each model simulation performed for 1.a-e, plot the cumulative number of prey eaten by the predator with time.

- **a (15 pts).** Test the hypothesis that reductions in the zone of detection decrease the number of prey eaten by a squared factor. For example, try predator.r=1 and then while keeping all other parameters the same set predator.r=0.5 and see if the number of prey eaten drops by 1/4.
- **b (15 pts).** Evaluate how changes in velocity modify the scaling relationship.
- **c (15 pts).** Modify the diffusion coefficient of prey, does it modify the consumption rate of predators?
- **d (15 pts).** Change the density of prey (don't increase it beyond 50 or so) - how does prey consumption change.
- **e (15 pts).** Set `info.replenish_prey=0`. You should find that once prey are eaten they do not regenerate, how does that change the consumption rate over time?

1.3 Lions and tigers and handling time, oh my!

Now, let's see what happens to the predator when it takes time to handle and consume prey. Restore all parameters to their original values and set

```
predator.handling_time=0.5;
```

This will make the predator have to wait 0.5 seconds for each prey it consumes. The model will continue to show which prey has been eaten (via a blue circle), even as the prey moves on. You should interpret this as a captured prey element that is now moving with the predator.

Exercises 2 (25 pts): How does the total number of prey consumed compare to the prior case (original parameter except for handling time)? Can you set parameters such that predators are essentially handling-time limited (hint: set detection and fraction consumed to be nearly 1 and also increase handling time)? In which case, does your model agree with predictions (see hypothesized consumption rate below)?

So how handling time changes the functional response? In particular, when predators must take time to handle and consume their prey then the consumption rate is hypothesized to scale like

$$\text{consumption rate} = \frac{bNP}{1 + aN}$$

where b/a is equal to $1/T_h$ where T_h is the handling time [1].

[1] In preparing this lecture last year, I tweeted an animated gif of the simulation and was informed that a very similar model has just been used to assess the validity of functional responses. For your interest, it is available here: `https://peerj.com/preprints/2190/`

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